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Project: Inferential Statistics; Hypothesis Testing

Topic 1: Inferential Statistics: Population Mean: One Sample

Dataset: Adult Males Pulse Rates (beats per minute, bpm)

Source:

https://www.statcrunch.com/app/?dlim=comma&ft=true&dataurl=https%3a%2f%2ftdx.acs.pearson.com%2fGetPlayerFile.ashx%3fguid%3ddfa4d1e3-18ce-427f-b761-c443d11838de

Description: The dataset is the sample of the pulse rates of 40 adult males in beats per minute.

Row	Males
1	81
2	73
3	48
4	62
5	50
6	62
7	50
8	74
9	49
10	60
11	70
12	62
13	66
14	79
15	80
16	63
17	69
18	97
19	39
20	87
21	74
22	67
23	74
24	70
25	52
26	67
27	59
28	80
29	73
30	64
31	62
32	93
33	55
34	68
35	58
36	57
37	66
38	70
39	87
40	58

Refer to the accompanying data set and construct a 95% confidence interval estimate of the mean pulse rate of adult males. Compare the results.

$Sample\ size, n = 40$

Objectives:

(1.) Construct a 95% confidence interval estimate of the population mean pulse rates of adult males.

(2.) Interpret the confidence interval.

Parameter to estimate: Population Mean

Test: t-test

Reason for the Test: We were not given the population standard deviation.

Verify Requirements for the Test.

(1.) The sample is a simple random sample.

(2.) The sample size is at least 30, justifying normality based on the Central Limit Theorem.

Data Analysis Using R Language

```
# Define the significance level
alpha <- 0.05
df <- n - 1

# Calculate the critical t-value
tCrit <- qt(1 - alpha/2, df)

# Calculate the Margin of Error
marginError <- tCrit * (s / sqrt(n))

# Calculate lower and upper bounds
lowerBound <- meanSample - marginError
upperBound <- meanSample + marginError

# Display the results
cat("95% Confidence Interval: [",lowerBound, ",", upperBound, "]\n")
```

```
95% Confidence Interval: [ 28.88651 , 34.89349 ]
```

Interpretation: We are 95% confident that the population mean of pulse rates of adult males in bpm is between 28.88651 and 34.89349.

Topic 2: Hypothesis Testing: Testing a Claim About Population Mean

Dataset: Los Angeles Commute Times (in Minutes)

Source:

https://www.statcrunch.com/app/?dataurl=www.statcrunch.com%2Fbooksets%2Ftriola_stat14t%2Fdataset%2F31_Commute_Times.txt&dlim=tab&ft=true

Description: The dataset is a list of 200 Los Angeles commute times in minutes.

MyStatLab Data Set

StatCrunch Applets Edi

Row	commuteTimes
1	30
2	90
3	25
4	10
5	4
6	70
7	60
8	45
9	5
10	15

Row	commuteTimes
11	30
12	20
13	20
14	30
15	25
16	10
17	17
18	30
19	15
20	35
21	7
22	25
23	60
24	60
25	60
26	30
27	15
28	15
29	50
30	90
31	10
32	30
33	25
34	20
35	20
36	20
37	10
38	20
39	15
40	40
41	10
42	30
43	60
44	30
45	30
46	50
47	30
48	50
49	12
50	20

Row	commuteTimes
51	30
52	60
53	10
54	10
55	25
56	30
57	20
58	50
59	15
60	25
61	60
62	45
63	25
64	15
65	15
66	30
67	45
68	10
69	20
70	30
71	15
72	30
73	45
74	40
75	20
76	15
77	25
78	30
79	20
80	90
81	15
82	35
83	30
84	60
85	25
86	10
87	5
88	30
89	15
90	60

Row	commuteTimes
91	15
92	60
93	30
94	30
95	45
96	15
97	30
98	60
99	60
100	60
101	15
102	40
103	30
104	12
105	35
106	10
107	30
108	30
109	35
110	60
111	1
112	20
113	60
114	25
115	30
116	50
117	35
118	10
119	30
120	30
121	60
122	30
123	30
124	7
125	10
126	30
127	10
128	60
129	60
130	35

Row	commuteTimes
131	30
132	45
133	10
134	30
135	60
136	10
137	60
138	30
139	30
140	90
141	45
142	60
143	25
144	20
145	20
146	70
147	25
148	15
149	30
150	30
151	141
152	30
153	27
154	30
155	70
156	5
157	20
158	40
159	60
160	20
161	35
162	10
163	40
164	45
165	30
166	10
167	40
168	30
169	45
170	30

Row	commuteTimes
171	20
172	40
173	141
174	20
175	40
176	25
177	15
178	50
179	4
180	15
181	15
182	20
183	20
184	15
185	15
186	15
187	30
188	10
189	60
190	20
191	15
192	20
193	10
194	30
195	60
196	20
197	45
198	15
199	10
200	30

Variable: Los Angeles Commute Times

Question content area top Part 1Use the accompanying 200 Los Angeles commute times to test the claim that the mean Los Angeles commute time is less than 34 minutes. Assume that a simple random sample has been selected. Use a 0.10 significance level. Compare the sample mean to the claimed mean of 34 minutes. Is the difference between those two values statistically significant?

Objectives:

(1.) Test the claim that the mean Los Angeles commute time is less than 34 minutes using the Critical Value Method.

(2.) Test the claim that the mean Los Angeles commute time is less than 34 minutes using the Probability Value Method.

(3.) State the decision.

(4.) State the conclusion.

(5.) State the interpretation.

Verify Requirements for the Test.

(1.) The sample is a simple random sample.

(2.) The sample size is at least 30, justifying normality based on the Central Limit Theorem.

Null hypothesis:	$\mu = 34$ minutes
Alternative Hypothesis:	$\mu < 34$ minutes
Test:	One Sample, Left-tailed

Data Analysis using R language:

```r
# Load the dataset
data <- read.csv("CommuteTimes.csv")
commuteTimes <- data$commuteTimes

# Sample Statistics
sampleSize <- length(commuteTimes)          # Sample size
meanSample <- mean(commuteTimes)            # Sample mean
s <- sd(commuteTimes)                       # Sample standard deviation
meanPopulation <- 34                        # Hypothesized mean
alpha <- 0.10                               # Significance level

# Standard Error and Test Statistic (t-score)
se <- s / sqrt(sampleSize)
tStat <- (meanSample - meanPopulation) / se

# 1. Critical Value Method
tCrit <- qt(alpha, df = sampleSize - 1)
```

```r
# 2. Probability Value (P-value) Method
pVal <- pt(tStat, df = sampleSize - 1)

# 3. Confidence Interval Method (One-sided Upper Bound for alpha = 0.10)
# For a left-tailed test, we use the 1-alpha upper bound
upperBound <- meanSample + qt(1 - alpha, df = sampleSize - 1) * se

# Output the results
cat("Sample Mean:", meanSample, "\n")
cat("Test Statistic (tStat):", tStat, "\n")
cat("Critical Value (tCrit):", tCrit, "\n")
cat("P-value:", pVal, "\n")
cat("90% Upper Bound:", upperBound, "\n")

# Built-in R Function for verification
t.test(commuteTimes, mu = 34, alternative = "less", conf.level = 0.90)
```

```
Sample Mean: 31.89
Test Statistic (tStat): -1.385331
Critical Value (tCrit): -1.28582
P-value: 0.08375105
90% Upper Bound: 33.84843

	One Sample t-test

data:  commuteTimes
t = -1.3853, df = 199, p-value = 0.08375
alternative hypothesis: true mean is less than 34
90 percent confidence interval:
     -Inf 33.84843
sample estimates:
mean of x
    31.89
```

Critical Value Method:

$Critical\ z - value\ =\ -1.28582$

$Test\ Statistic\ =\ -1.385331$

$-1.385331\ <\ -1.28582$

The test statistic is less than the critical z-value. This means that it does not fall in the critical region.

Decision: I need to reject the null hypothesis.

Probability Value Method:

$probability\ value = 0.08375105$

$significance\ level = 0.1$

$0.08375105 < 0.1$

The probability value is less than the significance level.

Decision: I need to Reject the null hypothesis.

Conclusion: There is sufficient evidence to support the claim that the mean Los Angeles commute time is less than 34 minutes.

Interpretation: The population mean for the Los Angeles Commute Times is significantly less than 34 minutes.

References (MLA 9):

"StatCrunch." *Statcrunch.com*, 2026, www.statcrunch.com/app/?dataurl=www.statcrunch.com%2Fbooksets%2Ftriola_stat14t%2Fdataset%2F31_Commute_Times.txt&dlim=tab&ft=true. Accessed 29 Apr. 2026.

"StatCrunch." *Statcrunch.com*, 2026, www.statcrunch.com/app/?dlim=comma&ft=true&dataurl=https%3a%2f%2ftdx.acs.pearson.com%2fGetPlayerFile.ashx%3fguid%3ddfa4d1e3-18ce-427f-b761-c443d11838de. Accessed 29 Apr. 2026.